

Microprocessor Systems - CSU23021

Assignment #02 –Game to Teach Morse Code

Group 2 - Arijita Aggarwal · Marcel Czubak · Shipra Singh · William Blanco Morrissey

1. Workflow

1.1 Team Roles

Role	Member	Student ID	Email	Primary Responsibility
Project Workflow Owner	Marcel Czubak	24382965	czubakm@tcd.ie	Milestone planning, task allocation, GitLab Issues tracking
GitLab Workflow Owner	William Blanco Morrissey	24380299	blancomw@tcd.ie	Repository management, branch strategy, and merge request review
Project Documentation Owner	Shipra Singh	25370745	singhs20@tcd.ie	Report writing, Doxygen headers, document structure
Project Demonstration Owner	Arijita Aggarwal	22304992	araggarw@tcd.ie	Demo video, WS2812 LED integration, innovative features

1.2 Milestones

- **Milestone 1 (by 21 March 2026):** Hardware abstraction complete - GPIO ISR operational, dot/dash classification working in assembly, GP25 LED toggling, initial CMakeLists configured.
- **Milestone 2 (by 1 April 2026):** Core game logic complete - difficulty selector, all four levels, live system, streak counting, end-of-level statistics, Watchdog integration.
- **Milestone 3 (by 8 April 2026):** Full integration - WS2812 PIO RGB LED, PWM buzzer (Feature #2), console visual timeline (Feature #1), demonstration video, report finalised.

1.3 Weekly Task Breakdown

Task	Owner	Status
GPIO ISR, dot/dash detection, Morse buffer (ASM)	Marcel / William	Complete
CMakeLists and project setup	All	Complete
Welcome banner and slow-print animation	Shipra	Complete
Morse lookup table and decode function	Arijita	Complete

Task	Owner	Status
Difficulty selector (Morse input)	Shipra	Complete
Single-character game rounds (Levels 1 & 2)	Marcel / William	Complete
Word-level game rounds (Levels 3 & 4)	William	Complete
Level scaffolding, streak, lives loop, stats	William	Complete
Watchdog timer integration	Shipra / William	Complete
WS2812 PIO RGB LED driver and colour logic	Arijita / Marcel	Complete
Feature #2: PWM buzzer driver + buzz_morse_sequence	Marcel	Complete
Feature #1: Console press-duration bar chart	William	Complete
Demonstration video (≤ 3 minutes)	Arijita	Complete

A team coordination meeting on 25 March 2026 reviewed outstanding work and redistributed tasks. Word-array and level 3/4 logic (William), watchdog fix (William), RGB LED blue idle state (Arijita), and further watchdog iteration (Shipra) were identified as the next priorities and completed within the following week.

2. GitLab Collaboration

2.1 Repository Structure and Branching Strategy

William Blanco Morrissey created the shared GitLab repository (MicroprocessorAssignment_2) and provisioned access for all members. The team used a feature-branch workflow: each member worked on named branches (e.g. feature/level3andlevel4, buzzer-extra, led-states, shipra-watchdog-implementation, arijita-blueled, feature/console-timeline) and submitted merge requests to main. William reviewed and approved all pull requests. Commit messages followed a short, descriptive present-tense convention.

2.2 Commit History

The four screenshots below show the complete commit log from the initial commit on 16 March 2026 to the final merge on 8 April 2026.

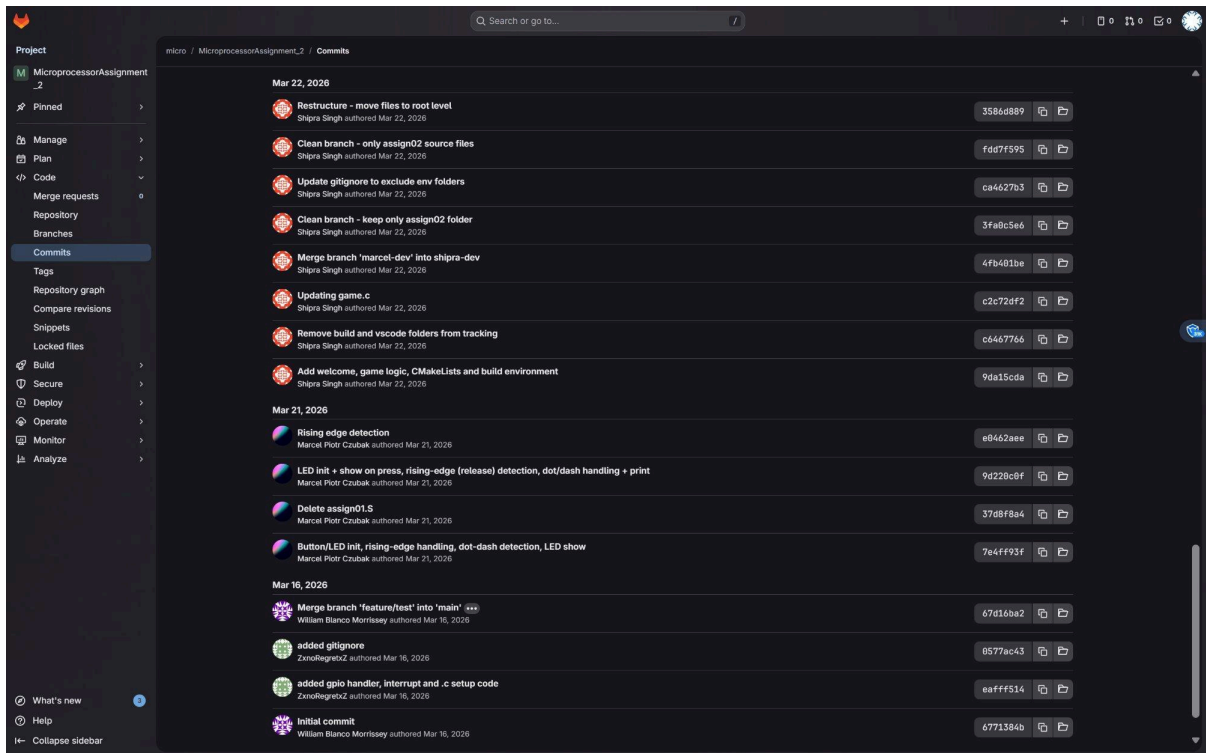


Figure 1 – Commits 16 Mar – 25 Mar 2026 (project setup and initial ASM/C foundations)

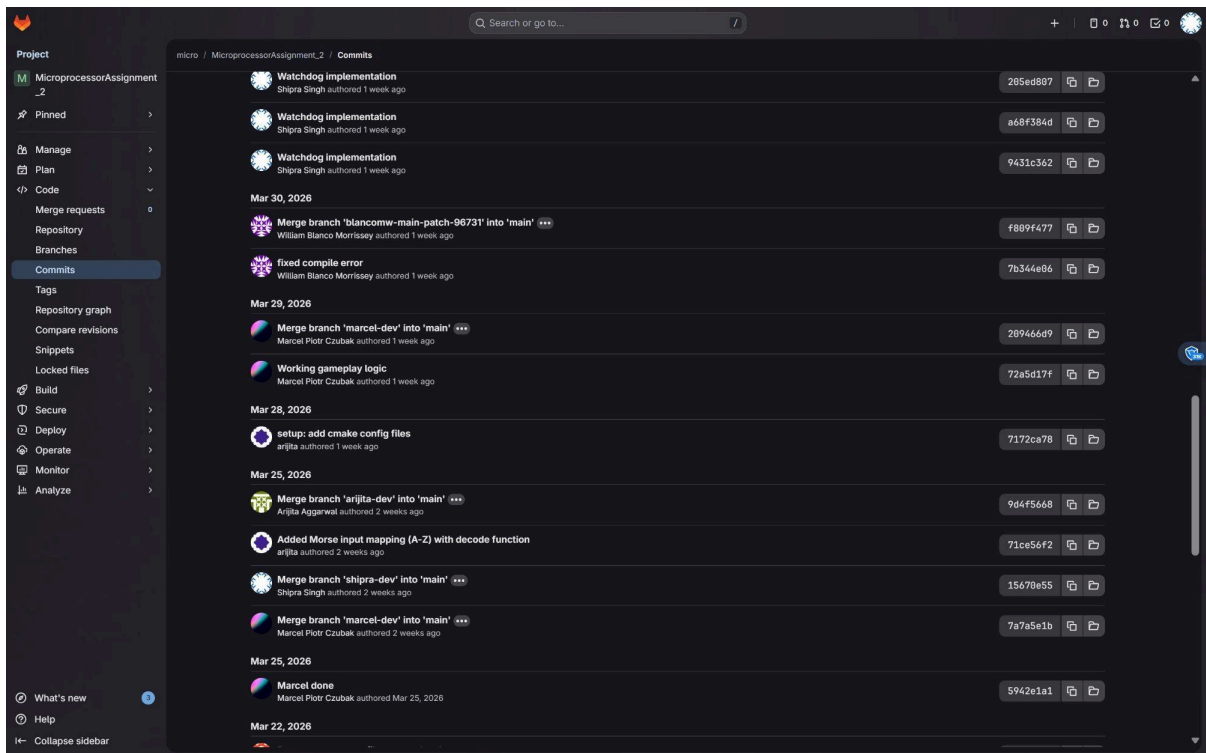


Figure 2 – Commits 25 Mar – 31 Mar 2026 (gameplay logic, watchdog, LED idle state)

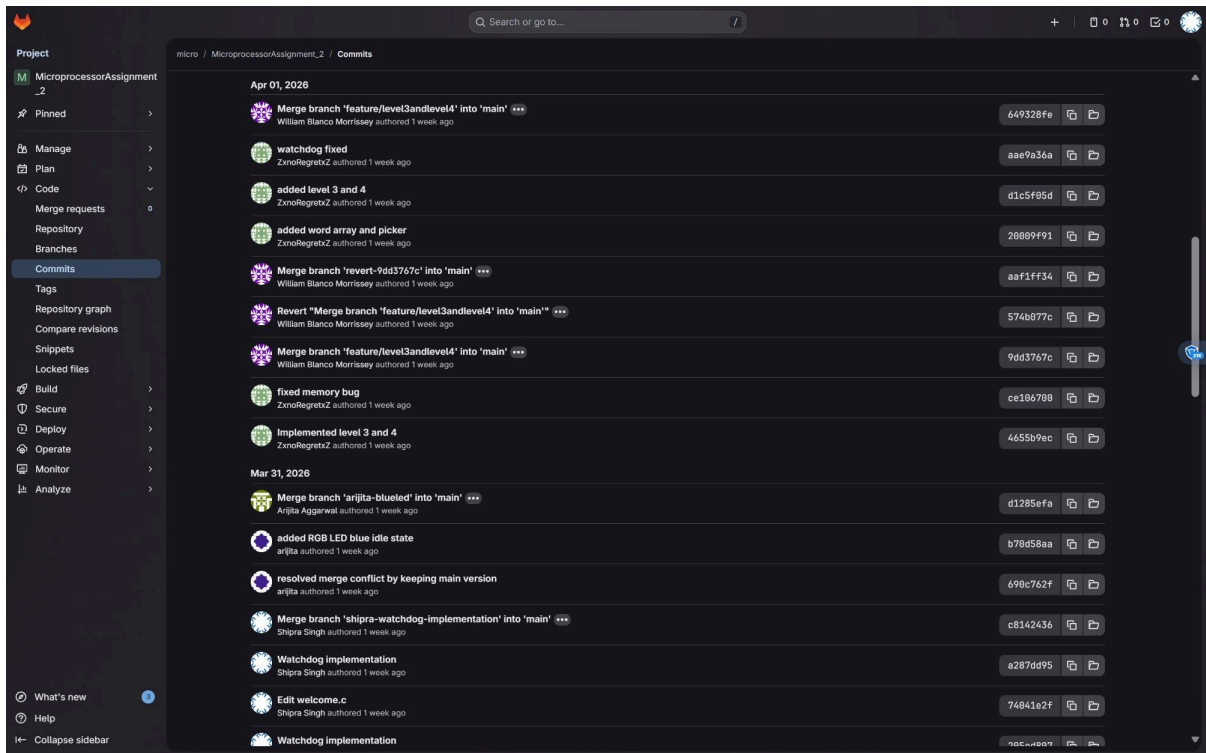


Figure 3 – Commits 31 Mar – 8 Apr 2026 (level 3/4, LED states, buzzer feature)

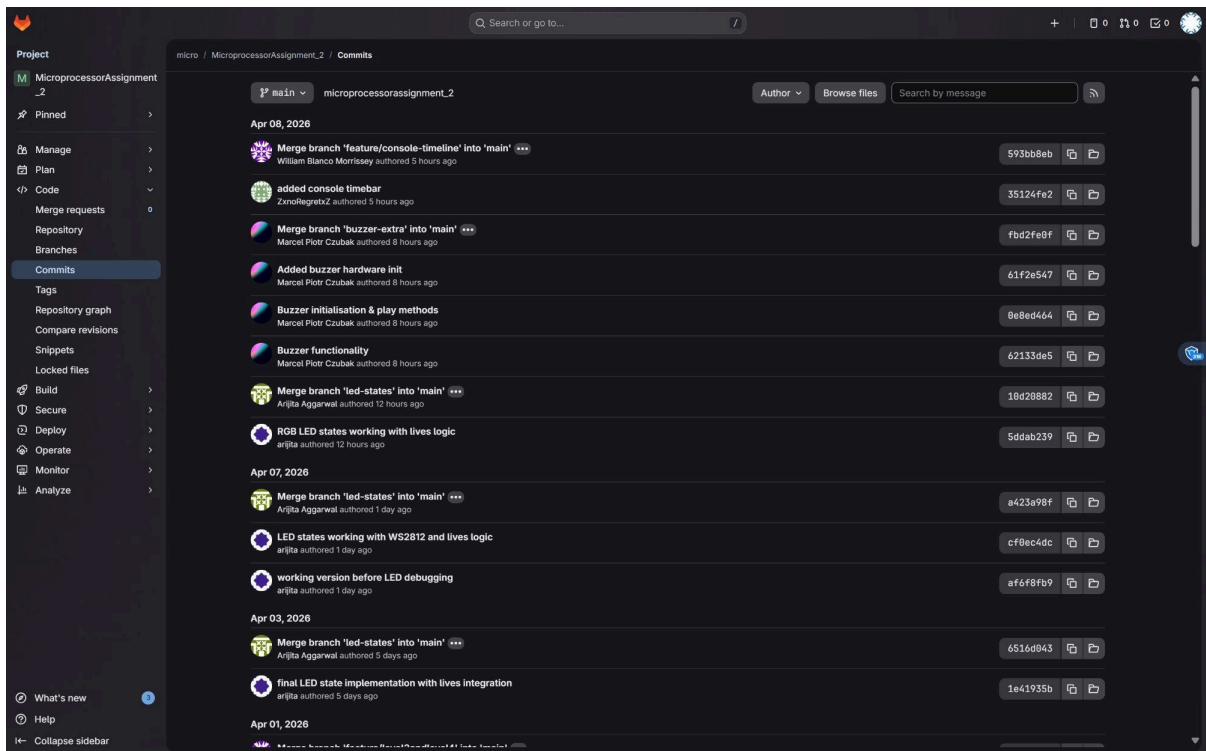


Figure 4 – Commits 8 Apr 2026 (final merges: console timeline, buzzer, LED states)

2.3 Commit Statistics by Member

Member	GitLab Username	Code Commits	Key Branches / Notable Commits
William Blanco Morrissey	blancomw	~9	feature/level3andlevel4, watchdog timer fix, feature/console-timeline; gpio handler, levels 3 & 4, console timebar (Feature #1)
Marcel Piotr Czubak	Marcel Piotr Czubak	~9	marcel-dev, buzzer-extra; dot/dash assembly, gameplay logic, buzzer init & play methods (Feature #2)
Shipra Singh	Shipra Singh	~10	shipra-dev, shipra-watchdog-implementation; welcome banner, game.c structure, watchdog, repo restructure
Arijita Aggarwal	arijita	~8	led-states, arijita-blueled, arijita-dev; Morse decode table, WS2812 LED states, RGB idle, cmake setup

Total commits across the project: approximately 50 (including merge commits). All four members made regular, active contributions across the full five-week period.

3. Individual Contributions

The following summaries document each team member's specific contributions to the codebase and project management, consistent with the GitLab commit history.

Marcel Czubak - Project Workflow Owner

- Led milestone planning, task tracking, and team coordination throughout the project via GitLab Issues.
- Implemented initial ARM assembly GPIO infrastructure: button/LED init, falling and rising-edge handling, dot/dash classification, GP25 LED toggle.
- Implemented working gameplay logic, integrating the C game loop.
- **Implemented Feature #2 (PWM buzzer):** buzzer_init, buzzer_play in assign02.c and buzz_morse_sequence in game.c, including the dual-frequency correct/incorrect scheme.

William Blanco Morrissey - GitLab Workflow Owner

- Created and maintained the shared GitLab repository; managed all merge requests and branch hygiene.
- Implemented initial GPIO handler, interrupt setup, and .c entry-point code.
- Implemented Levels 3 and 4 (word-level rounds): word array pools, pick_word, run_word_round.
- **Implemented Feature #1 (console visual timeline):** press-duration bar chart inside wait_for_input in game.c.
- Fixed compile errors and memory bugs during integration.

Shipra Singh - Project Documentation Owner

- Authored and compiled this project report, including the document template and structure.
- Set up the initial project repository structure: welcome.c, game.c, and CMakeLists.

- Implemented `print_welcome` in `welcome.c`, including ASCII art and the animated `print_slow` typing effect.
- Implemented `select_difficulty` and the initial single-character round logic (`run_single_char_round`).
- Integrated `watchdog_enable` and `watchdog_update` calls throughout the game loop.
- Restructured repository to remove build artefacts.

Arijita Aggarwal - Project Demonstration Owner

- Produced the project demonstration video (3 minutes) showcasing all four levels, progression, lives, LED feedback, and both innovative features.
- Added initial CMakeLists and cmake configuration files.
- Implemented the Morse lookup table and `decode_morse` function.
- Implemented the full WS2812 PIO-driven RGB LED colour logic integrated with the lives system.
- Added the Blue LED idle state to the startup sequence.